

ZELL AM SEE, Austria

WMO: 111440

Lat: 47.3267N

Lon: 12.7953E

Elev: 770

StdP: 92.41

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.9	-11.6	-16.0	1.0	-13.1	-13.7	1.3	-10.7	5.9	0.7	4.9	0.7	0.8	40	0.524

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	29.3	18.6	27.2	17.8	25.3	17.1	19.8	26.8	18.8	25.1	17.9	23.6	2.6	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.3	13.6	22.0	16.6	13.0	21.1	15.9	12.4	20.1	59.9	26.9	56.6	25.3	53.7	23.8	22.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.1	5.0	4.2	-17.3	32.5	3.8	1.8	-20.1	33.8	-22.3	34.9	-24.5	35.9	-27.3	37.2		
(5)			-17.9	21.5	4.1	1.1	-20.9	22.3	-23.3	22.9	-25.6	23.5	-28.7	24.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.2	-3.5	-1.0	3.9	8.7	13.3	16.4	18.2	17.9	13.7	9.2	3.0	-1.8	(6)
(7)		DBStd	8.48	4.33	4.13	3.94	4.06	3.64	3.98	3.55	3.30	3.45	3.70	3.96	3.90	(7)
(8)		HDD10.0	1650	417	308	192	71	12	3	0	1	8	59	212	367	(8)
(9)		HDD18.3	3799	676	541	448	289	158	81	47	48	142	283	459	626	(9)
(10)		CDD10.0	999	0	0	2	31	115	196	254	244	119	35	2	0	(10)
(11)		CDD18.3	107	0	0	0	0	3	24	42	34	3	0	0	0	(11)
(12)		CDH23.3	889	0	0	0	4	60	226	326	248	25	1	0	0	(12)
(13)		CDH26.7	228	0	0	0	1	9	59	94	63	2	0	0	0	(13)
(14)	Wind	WSAvg	1.5	1.1	1.2	1.6	1.9	1.9	1.8	1.7	1.5	1.4	1.3	1.1	(14)	
(15)	Precipitation	PrecAvg	1421	76	75	99	89	136	174	192	182	132	98	88	81	(15)
(16)		PrecMax	1604	186	200	226	152	200	273	293	286	244	173	182	232	(16)
(17)		PrecMin	1182	7	18	38	15	34	104	87	116	54	22	4	20	(17)
(18)		PrecStd	111	45	46	50	40	44	42	52	43	45	41	40	44	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.0	10.8	18.1	23.2	28.0	31.2	31.9	31.1	26.1	21.1	17.1	8.1	(19)
(20)			MCWB	5.6	5.5	9.5	12.7	16.4	18.8	20.3	19.7	17.6	13.9	10.2	5.5	(20)
(21)		2%	DB	4.2	7.2	15.3	20.7	25.0	28.7	29.2	28.5	23.4	18.3	12.5	5.2	(21)
(22)			MCWB	2.8	3.7	8.2	11.8	15.1	18.0	18.8	18.7	16.1	13.1	8.4	3.8	(22)
(23)		5%	DB	2.7	5.2	12.6	18.2	22.7	26.2	27.0	26.1	21.2	16.2	10.0	3.8	(23)
(24)			MCWB	1.8	2.6	7.0	10.3	14.0	17.1	18.1	17.9	15.1	12.0	7.7	2.8	(24)
(25)		10%	DB	1.7	3.8	9.9	16.0	20.4	23.8	24.8	24.0	19.2	14.4	8.1	2.6	(25)
(26)			MCWB	1.0	1.9	5.9	9.5	13.1	16.2	17.3	17.1	14.3	11.2	6.6	1.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.5	5.8	10.3	13.7	17.4	20.3	21.1	21.2	18.3	15.2	10.7	6.0	(27)
(28)			MCDWB	7.5	9.4	16.5	21.5	24.7	28.3	29.4	28.9	24.2	19.1	15.0	7.5	(28)
(29)		2%	WB	3.0	4.2	8.7	12.2	15.9	18.9	19.8	19.7	17.0	13.7	9.1	4.2	(29)
(30)			MCDWB	3.9	6.7	14.2	19.4	23.2	26.3	26.8	26.5	21.9	17.1	11.7	5.1	(30)
(31)		5%	WB	1.9	3.0	7.5	11.0	14.7	17.9	18.8	18.6	15.9	12.7	7.9	2.8	(31)
(32)			MCDWB	2.4	4.6	11.8	17.2	20.9	24.2	24.8	24.5	20.0	15.5	9.9	3.5	(32)
(33)		10%	WB	1.1	2.1	6.1	9.8	13.7	16.9	17.9	17.6	14.9	11.6	6.6	1.9	(33)
(34)			MCDWB	1.5	3.4	9.3	14.7	19.0	22.5	23.5	22.5	18.2	14.0	8.1	2.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.2	7.9	9.6	10.7	11.0	10.5	10.6	10.4	9.6	9.0	6.3	5.3	(35)
(36)			MCDBR	6.3	9.9	14.6	15.2	15.6	15.3	14.8	14.5	13.1	11.8	9.9	6.0	(36)
(37)			MCWBR	4.9	6.7	7.9	7.2	6.6	5.8	5.8	6.0	6.4	6.8	6.3	4.6	(37)
(38)		5% WB	MCDBR	5.7	8.5	13.5	14.0	14.0	13.8	13.2	13.4	11.9	10.9	8.9	5.6	(38)
(39)			MCWBR	4.6	6.0	7.6	6.9	6.2	5.5	5.6	6.0	6.1	6.5	6.0	4.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.276	0.306	0.351	0.394	0.409	0.416	0.415	0.407	0.370	0.340	0.308	0.277	(40)	
(41)		taud	2.372	2.284	2.233	2.216	2.239	2.265	2.271	2.312	2.390	2.445	2.465	2.432	(41)	
(42)		Ebn at Noon	815	855	862	857	856	850	845	836	837	809	771	773	(42)	
(43)		Edh at Noon	70	94	117	132	134	131	129	119	100	81	65	59	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.32	2.25	3.24	4.11	4.71	5.22	5.31	4.72	3.46	2.41	1.36	1.08	(44)	
(45)		RadStd	0.16	0.28	0.38	0.52	0.57	0.53	0.50	0.55	0.46	0.26	0.23	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.61	N/A	N/A	+1.06	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (91 neighbors)		+0.43	N/A	N/A	N/A	+0.40	+0.40	N/A	N/A	+69	+33

Nomenclature: See separate page